

## 0.1 A uniform Gaussian-corner theorem

Write

$$\varphi(t) = (2\pi)^{-1/2} e^{-t^2/2}, \quad r_z = -\Phi^{-1}(z), \quad 0 < z < 1/2.$$

The next elementary lemma records the normal-tail estimates used below.

**Lemma 1** (Uniform normal-tail coordinates). *For  $q > 0$ , put*

$$\eta(q) = \log \left\{ \frac{q\Phi(-q)}{\varphi(q)} \right\}.$$

*Then*

$$-\log(1 + q^{-2}) \leq \eta(q) \leq 0. \quad (1)$$

*Consequently,*

$$\log \Phi(-q) = -\frac{q^2}{2} - \log(\sqrt{2\pi}q) + \eta(q). \quad (2)$$

*Fix  $M < \infty$ . For  $r > 0$ ,  $z = \Phi(-r)$ , and  $0 < x \leq M$ , define*

$$q_r(x) = -\Phi^{-1}(zx).$$

*There are constants  $r_0(M)$  and  $C_M < \infty$  such that, for  $r \geq r_0(M)$ ,*

*1.*

$$q_r(x) \geq r - \frac{C_M}{r}, \quad 0 < x \leq M; \quad (3)$$

*2. whenever  $|\log x| \leq \sqrt{r}$ ,*

$$q_r(x) = r - \frac{\log x}{r} + \epsilon_r(x), \quad |\epsilon_r(x)| \leq \frac{C_M \{1 + |\log x| + (\log x)^2\}}{r^3}. \quad (4)$$

*All constants are deterministic.*

*Proof.* The classical Mills inequalities

$$\frac{q}{1+q^2} \varphi(q) \leq \Phi(-q) \leq \frac{\varphi(q)}{q}$$

give (1) and (2).

Let  $\ell = \log x$ ,  $q = q_r(x)$ . Since  $\Phi(-q) = x\Phi(-r)$ , (2) gives

$$\ell = -\frac{q^2 - r^2}{2} - \log \frac{q}{r} + \eta(q) - \eta(r). \quad (5)$$

For fixed  $M$ , Mills' inequalities imply  $\Phi(-r/2)/\Phi(-r) \rightarrow \infty$ , so  $q_r(M) \geq r/2$  for large  $r$ . By monotonicity,  $q_r(x) \geq q_r(M) \geq r/2$  for all  $x \leq M$ .

Suppose now that  $|\ell| \leq \sqrt{r}$ , and put

$$q_0 = r - \frac{\ell}{r}.$$

Then

$$\begin{aligned} -\frac{q_0^2 - r^2}{2} - \log \frac{q_0}{r} &= \ell - \frac{\ell^2}{2r^2} - \log \left(1 - \frac{\ell}{r^2}\right) \\ &= \ell + O\left(\frac{|\ell| + \ell^2}{r^2}\right). \end{aligned} \tag{6}$$

Moreover  $q_0 \geq r/2$  for large  $r$ , and (1) gives

$$|\eta(q_0) - \eta(r)| \leq \frac{C}{r^2}.$$

Thus

$$\left| \log \frac{\Phi(-q_0)}{\Phi(-r)} - \ell \right| \leq \frac{C_M \{1 + |\ell| + \ell^2\}}{r^2}.$$

The derivative of  $s \mapsto \log \Phi(-s)$  is

$$-\frac{\varphi(s)}{\Phi(-s)},$$

whose absolute value is at least  $s$  by the upper Mills inequality. The entire interval between  $q$  and  $q_0$  lies above  $r/3$  for large  $r$ . The mean-value theorem therefore gives

$$|q - q_0| \leq \frac{C_M \{1 + |\ell| + \ell^2\}}{r^3},$$

which is (4).

Taking  $x = M$  in (4), and using monotonicity  $q_r(x) \geq q_r(M)$ , proves (3).  $\square$

**Theorem 2** (Uniform Gaussian-corner asymptotic). *Let  $1 \leq d \leq d_0$ , and let*

$$X \sim N_d(\mu, \Sigma), \quad \Omega = \Sigma^{-1}, \quad U = \Phi(X)$$

*with  $\Phi$  applied coordinatewise. Let the tuple*

$$(d, \mu, \Sigma, D, a, w)$$

*range over a family satisfying the following uniform conditions.*

(G1) *There are constants*

$$0 < \omega_- \leq \omega_+ < \infty, \quad M_\mu < \infty, \quad \nu_0 > 0,$$

*such that*

$$\omega_- \leq \lambda_{\min}(\Omega) \leq \lambda_{\max}(\Omega) \leq \omega_+, \quad \|\mu\|_2 \leq M_\mu,$$

*and, with*

$$\nu = \Omega \mathbf{1}, \quad \kappa = \mathbf{1}^\top \Omega \mathbf{1}, \quad b = \mathbf{1}^\top \Omega \mu,$$

*one has*

$$\min_{1 \leq i \leq d} \nu_i \geq \nu_0.$$

(G2) There are constants

$$\eta_0 > 0, \quad M_D < \infty, \quad 0 < a_- \leq a_+ < \infty, \quad 0 < c_* \leq C_* < \infty, \quad d_* > 0$$

such that  $D : [0, 1]^d \rightarrow [0, \infty)$  satisfies

$$D(0) = 0, \quad \nabla D(0) = a = (a_1, \dots, a_d), \quad a_- \leq a_i \leq a_+,$$

$$\sup_{u \in [0, \eta_0]^d} \|\nabla^2 D(u)\|_{\text{op}} \leq M_D,$$

$$c_* \|u\|_1 \leq D(u) \leq C_* \|u\|_1, \quad u \in [0, \eta_0]^d,$$

and

$$\inf_{u \notin [0, \eta_0]^d} D(u) \geq d_*.$$

(G3) The weight  $w$  is positive and measurable, and there are  $0 < w_- \leq w_+ < \infty$  and  $L_w < \infty$  such that

$$w_- \leq w(0) \leq w_+, \quad |w(u) - w(0)| \leq L_w \|u\|_1, \quad u \in [0, \eta_0]^d.$$

The unweighted case corresponds to  $w \equiv 1$ .

Then, uniformly over this family, as  $z \downarrow 0$ ,

$$\begin{aligned} & \mathbb{E} [w\{\Phi(X)\} \mathbf{1}_{\{D(\Phi(X)) \leq z\}}] \\ &= w(0) C(\mu, \Sigma, a) z^\kappa \{\sqrt{2\pi} r_z\}^{\kappa-d} e^{-br_z} \{1 + O(r_z^{-1})\}, \end{aligned} \quad (7)$$

where

$$C(\mu, \Sigma, a) = |\Sigma|^{-1/2} e^{-\mu^\top \Omega \mu / 2} \frac{\prod_{i=1}^d \Gamma(\nu_i) a_i^{-\nu_i}}{\Gamma(\kappa + 1)}. \quad (8)$$

Equivalently, there are  $z_1 > 0$  and  $C < \infty$ , depending only on the constants in (G1)–(G3), such that

$$\sup_{\text{family}} r_z \left| \frac{\mathbb{E}[w\{\Phi(X)\} \mathbf{1}_{\{D(\Phi(X)) \leq z\}}]}{w(0) C(\mu, \Sigma, a) z^\kappa (\sqrt{2\pi} r_z)^{\kappa-d} e^{-br_z}} - 1 \right| \leq C$$

for  $0 < z \leq z_1$ .

In particular, if

$$H(z) = \mathbb{E} [w\{\Phi(X)\} \mathbf{1}_{\{D(\Phi(X)) \leq z\}}],$$

then there are uniform  $z_0, h_0, C_D > 0$  such that

$$H(z_0) \geq h_0, \quad H(2z) \leq C_D H(z), \quad 0 < z \leq z_0/2.$$

Thus the weighted deficit masses satisfy condition (P3).

*Proof.* We divide the proof into five steps.

*Step 1: density after corner rescaling.* The density of  $U = \Phi(X)$  is

$$f_U(u) = |\Sigma|^{-1/2} \exp \left\{ -\frac{1}{2} z(u)^\top (\Omega - I) z(u) + z(u)^\top \Omega \mu - \frac{1}{2} \mu^\top \Omega \mu \right\}, \quad (9)$$

where  $z_i(u) = \Phi^{-1}(u_i)$ .

Put  $r = r_z$ , and, for  $x_i > 0$ ,

$$q_i = -\Phi^{-1}(zx_i).$$

Thus  $z(zx) = -q$ . Define

$$A_z = |\Sigma|^{-1/2} e^{-\mu^\top \Omega \mu / 2} z^\kappa (\sqrt{2\pi} r)^{\kappa-d} e^{-br}, \quad (10)$$

and

$$g_z(x) = \frac{z^d f_U(zx)}{A_z}.$$

By (2),

$$\log z = -\frac{r^2}{2} - \log(\sqrt{2\pi} r) + \eta(r), \quad |\eta(r)| \leq r^{-2}.$$

Substitution into (9) gives the exact identity

$$\log g_z(x) = -\frac{1}{2} q^\top (\Omega - I) q - q^\top \Omega \mu + \frac{1}{2} (\kappa - d) r^2 + br + (d - \kappa) \eta(r). \quad (11)$$

*Step 2: a uniform integrable envelope near all coordinate faces.* Choose

$$0 < \vartheta < \frac{1}{2} \min\{1, \nu_0, \omega_-\}. \quad (12)$$

We prove that, for every fixed  $M < \infty$ ,

$$g_z(x) \leq C_M \prod_{i=1}^d x_i^{\vartheta-1}, \quad x \in (0, M]^d, \quad (13)$$

uniformly for small  $z$ .

For each  $i$ , (5) gives

$$\log x_i = -\frac{q_i^2 - r^2}{2} - \log \frac{q_i}{r} + \eta(q_i) - \eta(r).$$

Since  $q_i \geq r - C_M/r$  by Lemma 1, both  $\eta(q_i)$  and  $\eta(r)$  are  $O(r^{-2})$ , uniformly. Combining this identity with (11) yields

$$\begin{aligned} & \log g_z(x) - (\vartheta - 1) \sum_{i=1}^d \log x_i \\ &= -\frac{1}{2} \left[ q^\top (\Omega - \vartheta I) q - r^2 \mathbf{1}^\top (\Omega - \vartheta I) \mathbf{1} \right] - (q - r \mathbf{1})^\top \Omega \mu \\ & \quad + (\vartheta - 1) \sum_{i=1}^d \log \frac{q_i}{r} + O(r^{-2}). \end{aligned} \quad (14)$$

Write  $s = q - r \mathbf{1}$  and  $B = \Omega - \vartheta I$ . By (12),

$$\lambda_{\min}(B) \geq \omega_-/2, \quad (B \mathbf{1})_i = \nu_i - \vartheta \geq \nu_0/2.$$

Hence

$$-\frac{1}{2} \{q^\top B q - r^2 \mathbf{1}^\top B \mathbf{1}\} = -r (B \mathbf{1})^\top s - \frac{1}{2} s^\top B s.$$

The negative parts of the  $s_i$ 's are bounded by  $C/r$ , again by (3). Therefore

$$-r(B\mathbf{1})^\top s \leq -\frac{\nu_0 r}{2} \sum_i s_i^+ + C.$$

Also, by Young's inequality,

$$-\frac{1}{2}s^\top Bs - s^\top \Omega \mu \leq -\frac{\omega_-}{8}\|s\|_2^2 + C.$$

Finally, if  $q_i \geq r$ , then  $(\vartheta - 1) \log(q_i/r) \leq 0$ , whereas if  $q_i < r$ ,

$$\left| \log \frac{q_i}{r} \right| \leq Cr^{-2}.$$

Thus the right side of (14) is bounded above by a uniform constant. This proves (13).

In particular, on every fixed box,  $\prod_i x_i^{\vartheta-1}$  is integrable. Moreover,

$$\int_{\substack{x \in [0, M]^d \\ \min_i x_i < e^{-\sqrt{r}}}} \prod_{i=1}^d x_i^{\vartheta-1} dx \leq Ce^{-\vartheta\sqrt{r}} = o(r^{-1}). \quad (15)$$

This proves uniform integrable domination at every coordinate face, including intersections of several faces.

*Step 3: the density expansion away from the thin face region.* Let

$$\mathcal{E}_r(M) = \left\{ x \in (0, M]^d : \min_i x_i \geq e^{-\sqrt{r}} \right\}, \quad \ell_i = \log x_i.$$

On this set,

$$-\sqrt{r} \leq \ell_i \leq \log M.$$

Lemma 1 gives

$$q = r\mathbf{1} - \frac{\ell}{r} + e, \quad |e_i| \leq \frac{C_M(1 + |\ell_i| + \ell_i^2)}{r^3}.$$

Writing  $A = \Omega - I$  and expanding (11), one obtains

$$\log g_z(x) = \sum_{i=1}^d (\nu_i - 1)\ell_i + \frac{\ell^\top \Omega \mu}{r} - \frac{\ell^\top A \ell}{2r^2} + R_z(x), \quad (16)$$

where

$$|R_z(x)| \leq \frac{C_M}{r^2} \left\{ 1 + \sum_i |\ell_i| + \sum_i \ell_i^2 \right\}. \quad (17)$$

Indeed, the terms containing  $e$  are bounded by

$$Cr \sum_i |e_i| + C\|e\|_2 + C \frac{\|\ell\|_2 \|e\|_2}{r} + C\|e\|_2^2,$$

and (17) follows from  $d \leq d_0$ .

Consequently,

$$\left| \log \frac{g_z(x)}{\prod_i x_i^{\nu_i-1}} \right| \leq C_M \left[ \frac{1 + \sum_i |\log x_i|}{r} + \frac{1 + \sum_i (\log x_i)^2}{r^2} \right]. \quad (18)$$

The right side is  $O(r^{-1/2})$  on  $\mathcal{E}_r(M)$ . Hence, for large  $r$ ,

$$\left| g_z(x) - \prod_i x_i^{\nu_i-1} \right| \leq \frac{C_M}{r} \prod_i x_i^{\nu_i-1} \left\{ 1 + \sum_i |\log x_i| + \sum_i (\log x_i)^2 \right\}. \quad (19)$$

The parameters  $\nu_i$  range over a compact subset of  $(0, \infty)$ : their lower bound is  $\nu_0$ , while their upper bound follows from  $d \leq d_0$  and  $\|\Omega\|_{\text{op}} \leq \omega_+$ . Therefore

$$\sup_{\text{family}} \int_{[0, M]^d} \prod_i x_i^{\nu_i-1} \left\{ 1 + \sum_i |\log x_i| + \sum_i (\log x_i)^2 \right\} dx < \infty.$$

Integrating (19) and using (15) gives

$$\int_B g_z(x) dx = \int_B \prod_i x_i^{\nu_i-1} dx + O(r^{-1}) \quad (20)$$

uniformly for every measurable  $B \subseteq [0, M]^d$ . This is the required integrated  $O(r^{-1})$  remainder.

*Step 4: the moving nonlinear boundary.* For sufficiently small  $z$ , the event  $D(u) \leq z$  is contained in  $[0, \eta_0]^d$ , by the uniform gap  $d_*$ . On this event,

$$c_* \|u\|_1 \leq D(u) \leq z,$$

so after writing  $u = zx$ ,

$$\|x\|_1 \leq c_*^{-1}.$$

Thus all rescaled domains lie in a common compact box.

Taylor's theorem and the Hessian bound give, uniformly on that box,

$$\frac{D(zx)}{z} = a^\top x + z R_D(z, x), \quad |R_D(z, x)| \leq C. \quad (21)$$

Hence, for a uniform  $C_0 < \infty$ ,

$$\{x \geq 0 : a^\top x \leq 1 - C_0 z\} \subseteq \left\{ x \geq 0 : \frac{D(zx)}{z} \leq 1 \right\} \subseteq \{x \geq 0 : a^\top x \leq 1 + C_0 z\}. \quad (22)$$

Put

$$J(t) = \int_{\{x \geq 0 : a^\top x \leq t\}} \prod_i x_i^{\nu_i-1} dx.$$

The change of variables  $x = ty$  gives

$$J(t) = t^\kappa J(1),$$

where

$$J(1) = \frac{\prod_i \Gamma(\nu_i) a_i^{-\nu_i}}{\Gamma(\kappa + 1)}. \quad (23)$$

The parameters  $\nu_i, a_i, \kappa$  range over compact sets, so  $J(1)$  is uniformly bounded above and away from zero. The sandwich (22) therefore gives

$$\int_{\{D(zx) \leq z\}} \prod_i x_i^{\nu_i-1} dx = J(1) + O(z) \quad (24)$$

uniformly. This proves stability under the moving nonlinear boundary, including its intersections with the coordinate faces.

Combining (20) and (24),

$$\int_{\{D(zx) \leq z\}} g_z(x) dx = J(1) + O(r^{-1}), \quad (25)$$

because  $z = o(r^{-1})$ .

*Step 5: the weight and the final constant.* On the rescaled event,

$$\|zx\|_1 \leq z/c_*,$$

so (G3) gives

$$w(zx) = w(0) + O(z)$$

uniformly. The envelope (13) is integrable, and  $w(0)$  is bounded above and away from zero. Thus

$$\int_{\{D(zx) \leq z\}} w(zx) g_z(x) dx = w(0)J(1) + O(r^{-1}). \quad (26)$$

Multiplication by the normalizer  $A_z$  in (10), followed by (23), proves (7) and (8).

It remains to prove uniform doubling. By Lemma 1, applied with fixed  $x = 2$ ,

$$r_{2z} = r_z - \frac{\log 2}{r_z} + O(r_z^{-3}). \quad (27)$$

Therefore

$$\left(\frac{r_{2z}}{r_z}\right)^{\kappa-d} \exp\{-b(r_{2z} - r_z)\}$$

is uniformly bounded above and away from zero for small  $z$ . Since  $\kappa$  is uniformly bounded,

$$\frac{H(2z)}{H(z)} = 2^\kappa \left(\frac{r_{2z}}{r_z}\right)^{\kappa-d} e^{-b(r_{2z}-r_z)} \{1 + O(r_z^{-1})\}$$

is uniformly bounded. Choosing one sufficiently small  $z_0$  also makes the relative error in (7) smaller than  $1/2$ . The leading constant at this fixed  $z_0$  has a positive uniform lower bound, proving  $H(z_0) \geq h_0$ .  $\square$

## 0.2 Uniform verification for the conditional Gaussian families

Let

$$R_4 = (\rho^{|k-\ell|})_{1 \leq k, \ell \leq 4}, \quad \rho = \frac{1}{4}.$$

Its precision matrix is

$$Q = R_4^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} 1 & -\rho & 0 & 0 \\ -\rho & 1+\rho^2 & -\rho & 0 \\ 0 & -\rho & 1+\rho^2 & -\rho \\ 0 & 0 & -\rho & 1 \end{pmatrix}, \quad (28)$$

and

$$Q\mathbf{1} = \left(\frac{4}{5}, \frac{3}{5}, \frac{3}{5}, \frac{4}{5}\right)^\top. \quad (29)$$

**Corollary 3** (The corner theorem applies uniformly to A1–A3 and B1). *Fix  $0 < \varepsilon < 1/2$ . For every coordinate  $j$ , every conditioning value*

$$u \in I_{\varepsilon/2},$$

*and every model A1–A3 or B1, the conditional Gaussian family of the free active latent coordinates satisfies Theorem 2, with*

$$\nu_0 = \frac{3}{5}.$$

*Its exponents are*

$$\kappa_j = \begin{cases} 34/15, & j = 1, 4, \\ 41/15, & j = 2, 3, \\ 14/5 + \lambda_j^2/(1 - \lambda_j^2), & j > 4 \text{ in the Gaussian AR block,} \\ 14/5, & j \in \{5, \dots, 24\} \text{ in B1,} \end{cases} \quad (30)$$

where

$$\lambda_j = \rho^{j-4}.$$

In particular,

$$\frac{34}{15} \leq \kappa_j \leq \frac{43}{15}.$$

The B1 deficit mass is a weighted Gaussian-corner mass whose weight satisfies (G3) uniformly.

*Proof.* We first identify all conditional precision matrices.

If  $j \in \{1, 2, 3, 4\}$ , the conditional precision matrix of  $Z_{\{1,2,3,4\} \setminus \{j\}}$  given  $Z_j$  is the principal submatrix  $Q_{-j, -j}$ . For  $j = 1$ , its row-sum vector is

$$\left( \frac{13}{15}, \frac{3}{5}, \frac{4}{5} \right)^\top,$$

whose sum is  $34/15$ . For  $j = 2$ , it is

$$\left( \frac{16}{15}, \frac{13}{15}, \frac{4}{5} \right)^\top,$$

whose sum is  $41/15$ . The cases  $j = 3, 4$  are symmetric. Every component is at least  $3/5$ .

Now let  $j > 4$  belong to the original Gaussian AR block. Writing  $\lambda = \rho^{j-4}$ , the covariance between  $(Z_1, \dots, Z_4)$  and  $Z_j$  is

$$\lambda R_4 e_4.$$

The conditional covariance of the active block is therefore

$$\Sigma_\lambda = R_4 - \lambda^2 R_4 e_4 e_4^\top R_4.$$

The Sherman–Morrison identity gives

$$\Sigma_\lambda^{-1} = Q + \frac{\lambda^2}{1 - \lambda^2} e_4 e_4^\top. \quad (31)$$

Hence

$$\Sigma_\lambda^{-1} \mathbf{1} = \left( \frac{4}{5}, \frac{3}{5}, \frac{3}{5}, \frac{4}{5} + \frac{\lambda^2}{1 - \lambda^2} \right)^\top,$$



and

$$\kappa = \frac{14}{5} + \frac{\lambda^2}{1 - \lambda^2}.$$

Since  $0 \leq \lambda \leq \rho = 1/4$ ,

$$\frac{\lambda^2}{1 - \lambda^2} \leq \frac{1}{15},$$

which gives the upper bound  $43/15$ .

In B1, a proxy coordinate is independent of the active Gaussian block. Conditioning on that proxy leaves covariance  $R_4$ , so the precision is  $Q$  and  $\kappa = 14/5$ . A background coordinate  $j \geq 25$  gives the family (31), with the still smaller range  $\lambda \leq \rho^{21}$ .

The conditional means are also uniform. If  $j \leq 4$ , the mean is a fixed covariance vector times  $\Phi^{-1}(u)$ . If  $j > 4$  is in the AR block, it is

$$\lambda R_4 e_4 \Phi^{-1}(u).$$

Since  $u \in I_{\varepsilon/2}$ ,  $\Phi^{-1}(u)$  ranges over a bounded interval. The covariance families just displayed are compact and uniformly positive definite. Thus (G1) holds.

It remains to verify the deficit functions. Write

$$\frac{1}{\gamma(x)} = 2e^{\psi(x_1, \dots, x_4)},$$

where

$$\begin{aligned} \psi_{A1}(x) &= \psi_{B1}(x) = \sum_{\ell=1}^4 x_\ell, \\ \psi_{A2}(x) &= \frac{1}{2} \sum_{\ell=1}^4 x_\ell, \end{aligned}$$

and

$$\psi_{A3}(x) = 0.8 \sum_{\ell=1}^4 x_\ell + 0.20x_1x_2 + 0.15x_3x_4.$$

Fixing the conditioning coordinate and denoting the free active coordinates by  $y$ , let

$$\psi_0 = \psi|_{y=0}.$$

Then the reciprocal-index deficit is

$$D(y) = 2e^{\psi_0} \left[ e^{\psi(y) - \psi_0} - 1 \right]. \quad (32)$$

For A1 and B1, every free-coordinate derivative of  $\psi$  is one. For A2 it is  $1/2$ . For A3, every such derivative at the free corner lies in  $[0.8, 1]$ : a derivative is either  $0.8$ ,  $0.8 + 0.20u$ , or  $0.8 + 0.15u$ . Thus every component of  $\nabla D(0)$  is uniformly bounded above and away from zero. The Hessians are uniformly bounded, and strict coordinatewise monotonicity shows that  $D(y) = 0$  only at  $y = 0$ . Compactness of the remaining conditioning parameters gives the uniform two-sided corner comparison and the uniform gap outside a fixed corner neighbourhood. Hence (G2) holds.

For A1–A3 the weight is one. For B1, let

$$A(F) = \frac{\Phi(F) - 1/2}{2}, \quad |A(F)| \leq \frac{1}{4}.$$

The leading conditional tail coefficient is

$$c_B(x) = \mathbb{E} \left[ e^{A(F)/\gamma(x)} \mid U = x \right].$$

For the deficit mass conditional only on  $U_j = u$ , the tower property gives

$$\begin{aligned} & \mathbb{E} [c_B(U) \mathbf{1}_{\{D(U_{\mathcal{A}}) \leq z\}} \mid U_j = u] \\ &= \mathbb{E} \left[ e^{A(F)/\gamma(U)} \mathbf{1}_{\{D(U_{\mathcal{A}}) \leq z\}} \mid U_j = u \right]. \end{aligned} \quad (33)$$

After conditioning on the free active vector  $y$ , this is a Gaussian-corner integral with weight

$$w_{j,u}(y) = \mathbb{E} \left[ e^{h(y)A(F)} \mid U_j = u, U_{\mathcal{A} \setminus \{j\}} = y \right], \quad h(y) = \frac{1}{\gamma(y)}.$$

If  $j$  is not a proxy,  $F$  is standard normal and independent of the conditioning variables. If  $j$  is a proxy and  $t = \Phi^{-1}(u)$ , then

$$F \mid Z_j = t \sim N(0.7t, 1 - 0.7^2).$$

These means are uniformly bounded on  $I_{\varepsilon/2}$ .

For any Gaussian law of  $F$ , define

$$W(h) = \mathbb{E} e^{hA(F)}.$$

Differentiation under the expectation gives

$$\left| \frac{d}{dh} \log W(h) \right| = \left| \frac{\mathbb{E}[A(F)e^{hA(F)}]}{\mathbb{E}e^{hA(F)}} \right| \leq \frac{1}{4}.$$

In B1,

$$2 \leq h(y) \leq 2e^4,$$

and  $h$  has a uniform bounded derivative in the active PIT coordinates. Therefore

$$|\log w_{j,u}(y) - \log w_{j,u}(0)| \leq C\|y\|_1.$$

Also

$$e^{-e^4/2} \leq w_{j,u}(0) \leq e^{e^4/2}.$$

It follows that

$$|w_{j,u}(y) - w_{j,u}(0)| \leq C\|y\|_1$$

uniformly in  $j, u$ . This is (G3), and the weighted conclusion of Theorem 2 applies.  $\square$

### 0.3 Exact primitive tail representation for B1

We now prove the horizontal-coupling assertion with explicit constants.

Let

$$\lambda = 0.7, \quad \sigma^2 = 1 - \lambda^2 = 0.51, \quad \mathcal{P} = \{5, \dots, 24\}, \quad q = |\mathcal{P}| = 20.$$

The proxy coordinates are

$$Z_\ell = \lambda F + \sigma \epsilon_\ell, \quad \ell \in \mathcal{P},$$

where  $F, \epsilon_5, \dots, \epsilon_{24}$  are independent standard normals. The proxy block is independent of the base Gaussian AR block.

The B1 response is generated by

$$Y = S^{\gamma(U)} \exp \left\{ A(F) - \frac{1}{2S} \right\}, \quad S = V^{-1}, \quad V \sim \text{Unif}(0, 1), \quad (34)$$

where

$$A(f) = \frac{\Phi(f) - 1/2}{2}.$$

Put

$$h(x) = \frac{1}{\gamma(x)} = 2 \exp \left( \sum_{a=1}^4 x_a \right), \quad 2 \leq h(x) \leq h_+ := 2e^4.$$

Conditioning on the complete observed Gaussian vector  $Z = z$ , the posterior law of  $F$  is

$$F \mid Z = z \sim N\{m(z), s_*^2\}, \quad (35)$$

where

$$s_*^2 = \frac{\sigma^2}{\sigma^2 + q\lambda^2}, \quad m(z) = \beta \sum_{\ell \in \mathcal{P}} z_\ell, \quad \beta = \frac{\lambda}{\sigma^2 + q\lambda^2}. \quad (36)$$

Let  $W_0$  denote the principal Lambert  $W$ -function and set

$$R(x) = \begin{cases} W_0(x)/x, & x > 0, \\ 1, & x = 0. \end{cases}$$

Since  $x = W_0(x)e^{W_0(x)}$ ,

$$R(x) = e^{-W_0(x)}. \quad (37)$$

For  $h \in [2, h_+]$ ,  $a \in [-1/4, 1/4]$ , and  $t \geq 1$ , define

$$x_t(h, a) = \frac{h}{2} e^{-h(t-a)} \quad (38)$$

and

$$G_t(h, a) = e^{ha} R\{x_t(h, a)\} = \exp[ha - W_0\{x_t(h, a)\}]. \quad (39)$$

Finally, for  $G \sim N(0, 1)$ , define

$$c(h, m) = \mathbb{E} \exp\{hA(m + s_*G)\}, \quad (40)$$

$$H_t(h, m) = \mathbb{E} G_t\{h, A(m + s_*G)\}. \quad (41)$$

**Lemma 4** (Exact B1 tail factorization). *For every observed  $x \in (0, 1)^p$ , let  $z = \Phi^{-1}(x)$  coordinate-wise and put*

$$c_B(x) = c\{h(x), m(z)\}, \quad 1 + r_B(t, x) = \frac{H_t\{h(x), m(z)\}}{c\{h(x), m(z)\}}.$$

*Then, for every  $t \geq 1$ ,*

$$\mathbb{P}(Y > e^t \mid U = x) = c_B(x) e^{-t/\gamma(x)} \{1 + r_B(t, x)\} \quad (42)$$

*exactly. Moreover,*

$$-e^{1/2-2t} \leq r_B(t, x) \leq 0, \quad (43)$$

*and, wherever the derivative is taken,*

$$|\partial_t r_B(t, x)| \leq h_+ e^{1/2-2t}. \quad (44)$$

*Thus B1 satisfies (P2), with a remainder and log-level derivative of order  $O(e^{-2t})$ , uniformly in  $p$  and  $x$ .*

*Proof.* Fix  $x, F$ , write  $g = \gamma(x)$ ,  $h = 1/g$ ,  $a = A(F)$ , and let  $p$  be the conditional survival probability at  $e^t$ . By monotonicity of the generator,  $p$  is the unique solution in  $(0, 1)$  of

$$e^t = p^{-g} e^{a-p/2}.$$

Equivalently,

$$p e^{hp/2} = e^{-h(t-a)}.$$

Multiplying by  $h/2$  and applying  $W_0$ ,

$$p = \frac{2}{h} W_0 \left\{ \frac{h}{2} e^{-h(t-a)} \right\}.$$

Using (37),

$$\begin{aligned} p &= e^{-h(t-a)} R \left\{ \frac{h}{2} e^{-h(t-a)} \right\} \\ &= e^{-ht} G_t(h, a). \end{aligned} \tag{45}$$

Averaging (45) under the posterior (35) gives

$$\mathbb{P}(Y > e^t \mid U = x) = e^{-h(x)t} H_t\{h(x), m(z)\},$$

which is (42).

For  $t \geq 1$ ,  $a \leq 1/4$ , and  $h \geq 2$ , put  $y = h(t - a)$ . Then  $y \geq 3/2$ , and the function  $h \mapsto (h/2)e^{-h(t-a)}$  is decreasing on  $[2, \infty)$ . Hence

$$0 \leq x_t(h, a) \leq e^{-2(t-1/4)} = e^{1/2-2t}. \tag{46}$$

Since  $0 \leq W_0(x) \leq x$ ,

$$1 - x \leq e^{-W_0(x)} \leq 1.$$

The quotient  $H_t/c$  is a weighted average of  $R(x_t) = e^{-W_0(x_t)}$ , with weights proportional to  $e^{hA(F)}$ . Therefore

$$1 - e^{1/2-2t} \leq \frac{H_t(h, m)}{c(h, m)} \leq 1,$$

which proves (43).

Finally,

$$\partial_t \log R\{x_t(h, a)\} = \frac{h W_0\{x_t(h, a)\}}{1 + W_0\{x_t(h, a)\}},$$

so

$$0 \leq \partial_t R\{x_t(h, a)\} \leq h x_t(h, a).$$

Averaging with the same fixed weights gives

$$0 \leq \partial_t r_B(t, x) \leq h_+ \sup_{h, a} x_t(h, a) \leq h_+ e^{1/2-2t}.$$

□

#### 0.4 Uniform horizontal coupling for B1

Let

$$T_\varepsilon = \Phi^{-1}(1 - \varepsilon/2), \quad L_\varepsilon = \frac{1}{\varphi(T_\varepsilon)}, \quad \varphi_0 = \varphi(0) = \frac{1}{\sqrt{2\pi}}.$$

Also put

$$a_0 = \frac{1}{4}, \quad a_1 = \sup_f |A'(f)| = \frac{\varphi_0}{2},$$

and

$$C_h = \frac{1}{4} + \frac{5}{4}e^{-3/2}. \quad (47)$$

**Proposition 5** (Uniform B1 horizontal coupling). *For every  $p \geq 25$ , every coordinate  $j \leq p$ , every*

$$u \in I_\varepsilon, \quad v \in I_{\varepsilon/2}, \quad |u - v| \leq \varepsilon/2,$$

*the conditional kernels  $K_j(u, \cdot)$  and  $K_j(v, \cdot)$  admit a coupling*

$$X_u \sim \mathcal{L}(U \mid U_j = u), \quad X_v \sim \mathcal{L}(U \mid U_j = v)$$

*such that, almost surely,*

$$\|X_v - X_u\|_\infty \leq \varphi_0 L_\varepsilon |v - u|, \quad (48)$$

*and*

$$\left| \frac{1}{\gamma(X_v)} - \frac{1}{\gamma(X_u)} \right| \leq L_D |v - u|, \quad (49)$$

*where*

$$L_D = 8e^4 \varphi_0 L_\varepsilon. \quad (50)$$

*For the exact factorization (42), one also has, almost surely and for every  $t \geq 1$ ,*

$$\begin{aligned} & |\log c_B(X_v) - \log c_B(X_u)| \\ & + |\log\{1 + r_B(t, X_v)\} - \log\{1 + r_B(t, X_u)\}| \leq L_H |v - u|, \end{aligned} \quad (51)$$

*where the deterministic constant*

$$L_H = \left( \frac{3}{4} + \frac{5}{4}e^{-3/2} \right) L_D + 3e^4 \varphi_0 \lambda L_\varepsilon \quad (52)$$

*is independent of  $p, n, j, u, v, t$ .*

*Thus B1 satisfies condition (P4) with*

$$t_0 = 1, \quad \zeta_n = 0.$$

*Proof.* We first construct the coupling, then prove deterministic Lipschitz bounds for the two tail factors.

*Step 1: common-residual Gaussian coupling.* Let  $R_p$  denote the correlation matrix of the observed Gaussian vector  $Z = (Z_1, \dots, Z_p)$ . It has two independent blocks:

- on

$$\begin{aligned} \mathcal{B} &= \{1, 2, 3, 4, 25, \dots, p\}, \\ (R_p)_{k\ell} &= \rho^{|k-\ell|}; \end{aligned}$$

- on the proxy block  $\mathcal{P}$ ,

$$(R_p)_{kk} = 1, \quad (R_p)_{k\ell} = \lambda^2 \quad (k \neq \ell);$$

- correlations between  $\mathcal{B}$  and  $\mathcal{P}$  are zero.

Put

$$t_u = \Phi^{-1}(u), \quad t_v = \Phi^{-1}(v).$$

Let  $r_j = (R_p)_{-j,j}$ , and let

$$\eta_j \sim N\left(0, (R_p)_{-j,-j} - r_j r_j^\top\right).$$

Using the same  $\eta_j$ , define

$$\begin{aligned} Z_j^{(u)} &= t_u, & Z_{-j}^{(u)} &= r_j t_u + \eta_j, \\ Z_j^{(v)} &= t_v, & Z_{-j}^{(v)} &= r_j t_v + \eta_j. \end{aligned}$$

These are the correct conditional Gaussian laws given  $Z_j = t_u$  and  $Z_j = t_v$ . Coordinatewise,

$$Z_k^{(v)} - Z_k^{(u)} = (R_p)_{kj}(t_v - t_u). \quad (53)$$

Every correlation has absolute value at most one. Since  $u, v \in I_{\varepsilon/2}$ , the inverse-function theorem gives

$$|t_v - t_u| \leq L_\varepsilon |v - u|. \quad (54)$$

Consequently,

$$\|Z^{(v)} - Z^{(u)}\|_\infty \leq L_\varepsilon |v - u|.$$

Applying the global Lipschitz bound  $|\Phi(s) - \Phi(t)| \leq \varphi_0 |s - t|$  proves (48) for

$$X_u = \Phi\{Z^{(u)}\}, \quad X_v = \Phi\{Z^{(v)}\}.$$

*Step 2: reciprocal-index constant.* Writing  $z_a$  for the first four latent coordinates,

$$h(z) = \frac{1}{\gamma\{\Phi(z)\}} = 2 \exp \left\{ \sum_{a=1}^4 \Phi(z_a) \right\}.$$

For  $a \leq 4$ ,

$$\left| \frac{\partial h}{\partial z_a} \right| = h(z) \varphi(z_a) \leq 2e^4 \varphi_0.$$

Therefore

$$\begin{aligned} |h\{Z^{(v)}\} - h\{Z^{(u)}\}| &\leq 8e^4 \varphi_0 \|Z_{\{1,\dots,4\}}^{(v)} - Z_{\{1,\dots,4\}}^{(u)}\|_\infty \\ &\leq 8e^4 \varphi_0 L_\varepsilon |v - u|. \end{aligned} \quad (55)$$

This is (49)–(50).

*Step 3: movement of the posterior mean of  $F$ .* Recall

$$m(z) = \beta \sum_{\ell \in \mathcal{P}} z_\ell, \quad \beta = \frac{\lambda}{\sigma^2 + q\lambda^2}.$$

Under the common-residual coupling,

$$m\{Z^{(v)}\} - m\{Z^{(u)}\} = \beta(t_v - t_u) \sum_{\ell \in \mathcal{P}} (R_p)_{\ell j}.$$

If  $j \notin \mathcal{P}$ , the sum is zero. If  $j \in \mathcal{P}$ , then

$$\sum_{\ell \in \mathcal{P}} (R_p)_{\ell j} = 1 + (q - 1)\lambda^2 = \sigma^2 + q\lambda^2.$$

Hence, in all cases,

$$|m\{Z^{(v)}\} - m\{Z^{(u)}\}| \leq \lambda|t_v - t_u| \leq \lambda L_\varepsilon |v - u|. \quad (56)$$

This is where the fact that the proxy-block size  $q = 20$  is fixed enters. In fact its contribution cancels exactly in the posterior coefficient.

*Step 4: derivatives of  $c(h, m)$ .* Write  $F_m = m + s_* G$ , with  $G \sim N(0, 1)$ . Since  $|A| \leq a_0$  and  $|A'| \leq a_1$ , differentiation under the expectation in (40) gives

$$\partial_h \log c(h, m) = \frac{\mathbb{E}[A(F_m)e^{hA(F_m)}]}{\mathbb{E}e^{hA(F_m)}}, \quad |\partial_h \log c(h, m)| \leq a_0, \quad (57)$$

and

$$\partial_m \log c(h, m) = \frac{\mathbb{E}[hA'(F_m)e^{hA(F_m)}]}{\mathbb{E}e^{hA(F_m)}}, \quad |\partial_m \log c(h, m)| \leq h_+ a_1. \quad (58)$$

Thus, for any two parameter pairs,

$$|\log c(h_2, m_2) - \log c(h_1, m_1)| \leq a_0|h_2 - h_1| + h_+ a_1|m_2 - m_1|. \quad (59)$$

*Step 5: derivatives of  $H_t(h, m)$ .* For  $G_t$  in (39), put

$$x = x_t(h, a), \quad W = W_0(x).$$

The identities

$$xW'_0(x) = \frac{W}{1+W}, \quad \partial_h \log x = \frac{1}{h} - (t - a), \quad \partial_a \log x = h$$

give

$$\partial_h \log G_t(h, a) = a - \frac{W}{1+W} \left\{ \frac{1}{h} - (t - a) \right\}, \quad (60)$$

and

$$\partial_a \log G_t(h, a) = \frac{h}{1+W}. \quad (61)$$

As in the proof of Lemma 4, let  $y = h(t - a) \geq 3/2$ . Since  $W \leq x = (h/2)e^{-y}$ ,

$$\begin{aligned} |\partial_h \log G_t(h, a)| &\leq a_0 + \frac{x}{h} + x(t - a) \\ &= a_0 + \frac{1}{2}e^{-y} + \frac{y}{2}e^{-y} \\ &\leq \frac{1}{4} + \frac{5}{4}e^{-3/2} = C_h. \end{aligned} \quad (62)$$

Also

$$|\partial_a \log G_t(h, a)| \leq h_+. \quad (63)$$

Because  $H_t(h, m)$  is the expectation of the positive function  $G_t\{h, A(F_m)\}$ , its logarithmic derivatives are weighted averages of the corresponding logarithmic derivatives of the integrand. Thus

$$|\partial_h \log H_t(h, m)| \leq C_h \quad (64)$$

and

$$|\partial_m \log H_t(h, m)| \leq h_+ a_1. \quad (65)$$

The bounds hold uniformly for every  $t \geq 1$ .

Since

$$\log\{1 + r_B(t, x)\} = \log H_t\{h(x), m(x)\} - \log c\{h(x), m(x)\},$$

we obtain

$$\begin{aligned} & |\log\{1 + r_B(t, x_2)\} - \log\{1 + r_B(t, x_1)\}| \\ & \leq (C_h + a_0)|h_2 - h_1| + 2h_+ a_1|m_2 - m_1|. \end{aligned} \quad (66)$$

Adding (59) gives

$$\begin{aligned} & |\log c_2 - \log c_1| + |\log(1 + r_2) - \log(1 + r_1)| \\ & \leq (C_h + 2a_0)|h_2 - h_1| + 3h_+ a_1|m_2 - m_1|. \end{aligned} \quad (67)$$

Now use

$$C_h + 2a_0 = \frac{3}{4} + \frac{5}{4}e^{-3/2}, \quad 3h_+ a_1 = 3e^4 \varphi_0,$$

together with (49) and (56). This gives

$$|\log c_B(X_v) - \log c_B(X_u)| + |\log\{1 + r_B(t, X_v)\} - \log\{1 + r_B(t, X_u)\}| \leq L_H |v - u|$$

with  $L_H$  exactly as in (52). No exceptional deterministic term is needed, so  $\zeta_n = 0$ .  $\square$

**Remark 6** (What has been repaired). *The preceding arguments make six points explicit.*

1. *The Gaussian density is dominated uniformly all the way to intersections of coordinate faces by  $\prod_i x_i^{\vartheta-1}$ .*
2. *The normal-quantile expansion is used only off a face region whose normalized mass is  $o(r_z^{-1})$ .*
3. *The pointwise density error need not be  $O(r_z^{-1})$ : the mean term is proportional to  $r_z^{-1} \sum_i |\log x_i|$ . Its integral is  $O(r_z^{-1})$ , which is the statement actually required.*
4. *The nonlinear boundary contributes only  $O(z)$ , because it is trapped between two homothetic simplices. Since  $z = o(r_z^{-1})$ , it is negligible at the announced order.*
5. *A weight  $w$  changes the leading constant to  $w(0)C(\mu, \Sigma, a)$ . Omitting  $w(0)$  would be incorrect.*
6. *For B1, the horizontal constants are independent of the ambient dimension. The common-residual coupling moves every latent Gaussian coordinate by at most  $L_\varepsilon |u - v|$ ; the only apparent sum over the twenty proxies cancels exactly in the posterior mean coefficient.*